

ActInf Textbook Group ~ Cohort 1 ~ Meeting 20 (Chapter 8, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 20 (Chapter 8, part 2) | 52:12

hello everyone it's October 14 2022
we're in
meeting 20. we're in our second
discussion of Chapter eight we have a
few
topics and points in chapter eight and
then maybe we can
preview some chapter nine and have a lot
of space for any other thoughts or
questions people are having so
we can jump to the questions and look at
the um
written chapter 8 questions or does
anyone want to
suggest a point for us to go to
last time we know that some of the areas
we wanted to pick up on
involved learning and hierarchical
modeling and hybrid modeling
one question that came up I think
here but maybe we can revisit is the
Birdsong model
is widely pointed at as an example of
how communication is modeled in active
inference or for active inference
entities
however it's quite literally singing
from the same hymn sheet and the turn
taking is based upon uh joint
expectation of that song being sung that
way
and then the actions that the birds can
take the kind of turn taking that they
enact
is

against related to whether it's like
well somebody has to say it is it going
to be me or is it going to be that
so how do you think we would broaden
that scope
of communication to account for
situations
where there isn't a pre-scripted song
how do other Frameworks for
communication
cybernetic ish or otherwise
deal with
the semantics of conversation
improvisation
open-endedness in conversation
uh actually I had a
somewhat more general question about uh
the bird songs and its relation to uh
this particular systems that they've
developed in the textbook uh you see
some researchers believe that I'm sure
you're much uh qualified to talk about
uh this than me but
there are lots of research going going
around about the territorialization uh
function of bird songs so if we take
that as uh if we take a birth song as uh
as its territorializing mechanism at
least one of uh it's territorializing
mechanism uh can we
somehow
put the dynamic I mean the dynamic
boundary of the markup blanket uh I
think would be affected uh from this
point of view and then
I'm not sure how that would translate
into this specific model if we take that
Dynamic nature of the mark of blanket

boundary

yeah I guess I would just build on that
and say that um the communication serves
a purpose and so
um I think a model shouldn't incorporate
what that purpose is
and how it's served by the communication
and then whether it happens to be you
know syncing up with some predefined
pattern or or whatever is is a is a
follow-on consideration
the generative model in the singing from
the same hymn sheet
is like is very much like the sheet
music example
provided in the textbook
there's a known sheet music script
such that even when there's like a error
in the playing of the music
still the sheet music can be heard
so
just coarsely
what generative models
um I think it's pretty straightforward
to see how a generative model could have
the notation or the sheet music
and then use the sheet music as its
prior
to kind of converge reality
to what extent is communication that way
is it totally disparate and differently
structured or is it a little bit like a
hybrid like you're doing inference on
the sheet music that they're about to
say and then you use your inference
about what they're about to say to
converge what they do say
well you ask a good question which is

well what are some alternative

Frameworks and um

I I know I I know I know some or should

know some and I they're not coming to

mind what comes to mind uh most

immediately in this example is a phase

lock loop in electronics which is simply

a way of synchronizing one circuit's

oscillation with another

and um this is like a phase lock loop

with a bigger vocabulary

so figure out where we are in the in the

pattern and then

um

synchronize with it

and um

but to the answer the question of what

other Frameworks

um

I mean there's

this whole field of

um conversation analysis

um which

um well it feels actually called CA

conversation analysis and they have

identified patterns of

um how people communicate

through a certain

structure of I guess we call them

rhetorical devices for achieving the

purpose of the conversation so the

purpose of the purpose

of the communication is kind of Bedrock

but um and there are multiple purposes

some of which are instrumental some are

which are you know social

um

and then they have identified patterns

for the way human conversation tends to
proceed which are culturally influenced
you know there's some cross-cultural
aspects but then there's certain
different cultures that have different
takes on it so they identify things like
okay now we're in the conversation
opening phase now we're in a phase of
turn where we're exchanging information
and we have turn taking now we have
Notions of
um epistemic leadership or epistemic
authority
um you know who's who's the person who's
asking who's the person who's answering
or taking those uh those roles in the
conversation and and there's certain
rhetorical stances that are taken then
we have so
um that's a pretty well worked out
um I I don't I guess even formalized
system because they you know there's
just there are um you know well
identified patterns um it's it's not
strictly worked out because it seems
like every domain a conversation analyst
goes to and every every group has their
own kind of flavor of it but it is kind
of um
uh it it does have this aspect of
synchronizing
with respect to the underlying known
patterns of each of the parties in a
conversation
the bird song is also oh Ali please
uh sorry adding to what Eric just
mentioned uh that reminds me reminded me
of the uh a much mechanism on a much

much smaller scale
namely the scale of cells and so on uh
and uh particularly the model proposed
by Stuart Kaufman because
um he talks about the
pluripotent nature of cells and how they
communicate with each other uh in the
context of phenomena of induction
biological induction and uh I mean uh in
the embryological development stage of
the cells uh almost all cells are
pluripotent and they're capable of
becoming any of the different types of
cells which characterize the adult
individual so
um that also the uh from Kaufman's
model perspective
uh these kind of uh inductive signals or
so to speak communicative signals
between cells can act as a kind of
non-specific stimulus
which switches cell among a variety of
internally available stable States uh so
uh I mean a cell
or let's put it this way the basic idea
in this model is that the regulatory
genes within a cell form a complex
Network in which genes uh interacting by
their products can turn one another on
or off you see so uh yeah these patterns
exhibit the kind of homeostatic
stability associated with attractors of
the systems so uh I think that
translates perfectly into this scale as
well this is small
yeah the
the mapping of communication as
generalized synchrony

may be compatible with a variety of
communication mechanisms
and so it abstracts us from the media
of communication per se
because we can talk about the
informational synchronizations
between for example the two internal
states of the two entities
or multilateral synchronizations with
environmental States or so on
um
like thinking about
um the basic mechanics Bayesian physics
the synchronizing metronomes
they don't have a hem sheet they don't
have sheet music
their embodiments
and coupling
um
enables
synchronization
now one thing I might know you know you
I think you asked a good question here
which is you know how would you transfer
translate something like um
um conversation analysis
you know a human conversation into an
active inference model so
um but that that um
kind of reminds me that the continuous
valued
um models
because they're limited by these by what
you can do in these differential
equations
they seem to be fairly simple
whereas if I were to do something like
thinking about human conversation in

terms of active inference I would want
more something more a very highly High
dimensional or multi-state type of
system where I can

um

have more complex

um patterns

and they may not make these you know

Loops these Volterra Loops or something
called Volterra is that right or is this

is Lawrence

something Dynamics yeah

yeah so it may not make those continuous

Loops but I that you'd still get

um

um you know good structured patterns out
of discrete

um systems so I've I'd kind of be

curious I think this chapter would be

because if it what had if it had a um uh

something about communication and

synchrony pattern synchrony but in terms
of discrete value

um active inference models as a

continuous valued ones

yeah

yeah very interesting like

that's what I was wondering which is why

is communication it is

um

continuous time models are argued to be

a natural fit for motor control

our communication

situations Naturally Fit for one or the

other

seems like I know you could imagine it

either way and then also like

um the synchronization means the

internal state of one creature should
come to resolve the uh resemble internal
states of another A Primitive kind of
theory of mind I think
um not as a um uh
fatal critique
but these examples put the rabbit in the
Hat
a little bit too keenly like the
planning example we explored wasn't
truly planning it was like a
parameterized
um Subspace
where single step optimization
emulated planning now one could argue
that they believe that that's actually
like a better approach than than true
explicit planning but that's not what
was on the um cover and then similarly
theory of mind though qualitatively may
be a framing for communication
we're talking still about the bird song
where they have the same shared a priori
generative model
so this doesn't
um in other words theory of mind is is
um is axiomatically and implicitly
embedded within this model
rather than active inference being used
as uh in a priori first principles to
generate some realized computational
theory of mind
I feel like yeah yeah it's like um you
know another aspect of this the The
Continuous solute formulations is
there's really not that many degrees of
freedom
but when you're communicating you know

anything other than a fixed song there's
a lot of degrees of freedom and um
that's why we have symbol systems but
simple systems is where you would use a
discrete model framework for not a
continuous value

okay

let's talk about hybrid

models discrete and continuous models
and learning and updating so these are
two features chapter eight

is

um

within a continuous time framework

just like chapter seven

layered in

these kind of patterns or archetypes of
discrete time models

in chapter 8 analogously

the most

kernel continuous time formalism is

introduced with movement control

and then we're going to see some kind of
extension extensions or

um

patterns

like design patterns that include

continuous time

this is the message passing

okay so missing line here

but we have

here this top half

is like the bottom of figure 4.3 or at

the top I guess discrete time

so we have B Matrix as a transition

Matrix we have discrete time

and we have

um yeah it's discrete time t plus one T

minus one but within each

time point

we see the continuous time you know e

whatever we want to call this

motif

where the B transition Matrix and

discrete time is

functionally substituted

with a derivative map

such that rather than actually uh retro

or forecasting

State values through time this is much

more like a Taylor series approximation

where extrapolation from the presence

under the auspices of slower changing

causes V which play a functional role of

policies

higher derivatives capture more and more

variants further and further from

the origin

what is the ETA

doing

it's playing some functional role to G

again allowing for a line here

in the sense that it is

selecting policy

but it's not it's not policy that's

being selected or discussed this way but

these slower changing factors that

influence how the derivatives are made

but is

you know a free energy variable

let's just ask

okay

what do people think about this type of

hybrid modeling

um how does this

align with

the previous Decades of analog digital
signals engineering
and the interfaces the digitalization of
analog noise and signals and um so on
foreign

well um

I'll only offer one thought if it's all
right

um

so

one of the things that active inference
let you do is look at

a past trajectory

and figure out how to interpret it
in terms of your your variable your
state variables

um

so you can essentially you you look for
the

the probability distribution over
variables that optimizes the free energy
and that allows you to look back in time
and

um

reinterpret

um evidence because you get something
new and it you get you know in light of
new evidence and then some of the
earlier chapters they had this kind of
example and they had the example of the
of a dynamical a dynamic programming
kind of a a model for this so I would
expect that to be the case also in
communication

um systems and I guess I'm going back
again to communication where uh to some
extent you're able to reinterpret what
you thought you heard before

um you know we it's kind of rare in in
language but it happens there are Garden
Path sentences

um

but usually you know you most
communication systems try to be
relatively hierarchical in that once you
was once you interpret your continuous
value signal then you don't leave a lot
of ambiguity you kind of you know fix on
that and then you have to deal with it
later and sometimes you go back and do
repair

so I'm wondering if this figure 8.6 if
that allows that kind of Dynamics where
um you know you're kind of tracking
something in the continuous value level
you're tracking a signal why effectively
by making predictions of what you're
going to hear uh What You observe
um and then can is that able in this
framework to go and influence the symbol
level the uh the B you know the s
s variables above and how far back in
time is it able to do that

nice

that makes me think about phonemes down
here

and audio

and then here are the um
spellings so in a language or someone is
speaking unclearly or you're not super
familiar with the language

um

then you really are like doing inference
on the on the whole time series like
it's like that kind of
um retro casting which can include even

revisionism and and delusions and false
memories and all that now casting
including the tension between
representing things quote as they are
versus like
um having delusions in the present
and then anticipation which of course
hasn't even happened unless you have a
vision of time as you know
something that always was or
however
so
it's really
um
interesting that although this time T is
an index
we are doing inference across these
and maybe it would speak to some
um measure of information processing
and memory and anticipation
the extent to which
um
like meaning
is able to be
the closer meaning is to a critical
point like the more this graph structure
if someone says but I was just kidding
you would want that to apply to to be
like oh I didn't know I was in a
parenthesis and now that they were just
kidding about that
um
so it's actually like delineating those
structural markers
and then reinterpreting meaning
is
perhaps moving towards the kind of
linguistic

modeling

so yeah to come back to it

so this could be a waveform

the observables down here could could be

just a four-year analysis or or a pure

auditory

um waveform

yeah I think phonemes is a that's what

comes to mind also

okay

foreign

motor tasks

so selecting a Target location and

using motor Behavior functionally eyes

to foviate the Target

figure eight seven okay figure eight

seven

is a hybrid or mixed generative model

higher level is this is kind of like the

decision making motor active inference

Fusion

here we have a little bit of the italics

and bold

um

which we could dive into but don't need

to do so at this moment

so you know my recollection for last

week was that that corresponded to

um generative model versus process model

you think that might map again here I

know you didn't want to dive into it

yeah

all right

I believe it does let's just remind

ourselves where it came up earlier in

the chapter yeah it was the very

beginning

right here yeah with italics versus

process in bold model in italics
so eight seven
so now here
the um discrete model is on a time
period of 200 milliseconds
continuous model operates continuously
um I think this speaks to how
biological knowledge can be
encoded in the model structure
I mean a more comprehensive approach
would have done parameter sweeps for
what the optimal number of milliseconds
is to do a discrete model on
however one could say like well the
physiology that we're interested in is
accomplished with a gamma
EEG cycle the gamma EEG cycle is this
many milliseconds
so we're just going to treat
that as like a discrete Rhythm like one
can imagine if the discrete time were
one millisecond here
you would be losing computational
advantage and you'd be kind of like over
slicing
because if eye movement does happen on a
time scale of
hundreds of milliseconds
discretizing to one millisecond is going
to be too fine and discretizing to one
second is not going to enable multiple
actions to be taken within that frame
so time scale and spatial scale of a
system
might be able to be encoded
at the phase of its construction
or structured learning more generally
gaussian mixture model

and the hierarchical
yeah so before before we move on just
about the isocades
um I don't understand what is deemed by
an active inference model for this I
mean why not just say you got a
controller yeah it's a control Loop and
that's good enough
yes
in
um live stream number 50 which we're
just preparing for it's like very
classical cybernetic control Loop
homeostatic Dynamics
and um
it's interesting to
ask like again what makes the model Act
of inference
one could say that if there's a Markov
blanket but we're you know between every
one of these variables
it is a model of action and perception
like so many other models of action and
perception
so what is being shown here is it that
the single imperative the free that
energy-based imperatives
just speaking narrowly here
energy-based imperatives
can mediate
decision making in discrete continuous
Hybrid models
and that it's possible to have
a categorical decision-making model
in feedback
with A continuous ocular motor model
at the very least it places this type of
model

within the
domain and grammar of active inference
modeling
okay data clustering we may have touched
upon also a hierarchical gaussian filter
that I know some people have
worked on
okay let's look at some of the summary
advances in continuous time models
synthetic bird song we talked about this
ocular motor delays
conditioned reflex
that's okay so this is in the chapter
this is in the chapter
I don't know about this exact paper
perhaps but ocular Motors in the chapter
and smooth eye Pursuits
conditioned reflexes that is in this
chapter the blank the conditioned and
the unconditioned to stimuli
building on the Songbird model but the
Songbird model was in 2015.
false inference from sub-optimal prior
beliefs
can but prior beliefs be sub-optimal
um sorry uh actually McPherson has
talked about bird songs uh before 2015
in uh one of his papers I think from
2013 or 12 uh yeah it's it's from 2013
which is co-authored with uh his son I
guess Dominic aphriston so yeah I mean
2015 paper was not the first paper the
name of the paper is the is the free
energy formulation of Music generation
and perception thank you
thank you yeah
wow yeah good um find a reminder
interesting

looks very much like the textbook you

know opening quotes

I can see how this is relates here

researching too

okay

oh maybe even 2009 bird song chaotic

attractors wow you know

it's 10 10 plus years on

the bird song

paradigm

okay so um

deviations in perception

more eye movements

it's kind of fun like how we've talked

about babbling and that across different

domains like cicading

no there's the ocular motor cicade but

are there other kinds of cicadings

[Music]

um

for example I think that

Knowledge Management

niches facilitate semantic cicading yes

enabled by visual cicading as well

because you have to be like looking

around on the page but it's just like

what was chapter four

okay okay that's what we're talking

about chapter four

what's notation I think it's just sort

of like

the analog to scanning around a little

bit

conversational analysis that's you know

that's a very important topic is topic

control

and um or you know if you're uh

if you look at um any you know most

conversations have multiple threads

going

so it's like a Fugue in you know Bach

okay you know which voice is this are

you you know every note is in a voice

and the voice has got some topic and

they intertwine

in girdle escherbach

Prelude Aunt fugue

nice this is very um interesting I think

it starts to

[Music]

um

plant some seeds towards again semantic

cicading computational conversation

analysis and generation

adaptive knowledge environments

like visual accessibility is

one topic you know if somebody has um a

visual difference then making sure that

they can see it is the enabling system

but let's just assume that we're within

a space where we can talk about what is

being seen so then

um how does someone get a clear picture

of what we're talking about

that would be through some sort of

cicade

where you'd recognize

um more uncertainty at the periphery

and um like the fovea in this case is

paying attention to the conversation

and that's why like with a language

we're unfamiliar with we get a little

bit lost because like our circuit was

too slow we don't have the fluency to

cicade and so the video is moving too

fast

to track

so we're just kind of like randomly

listening and like trying to like

compute

but what if somebody were on a guided

cicade journey through their semantic

landscape

I mean just to take this a little bit

further um

this is also maybe uh you know it's a

critical element of

uh cognitive architecture which is the

ability to topic switch and you know in

Psychology and cognitive science we

there's this notion of executive control

and one of the fundamentals of executive

control is to be able to

topic switch so you're on some task you

get an interrupt you have to put that on

a stack deal with your interrupt then

when you're done and maybe the interrupt

is a goal it could be a sub goal it

could be some other topic that comes in

something you have to deal with then

when you're done you have to be able to

pull that back push that back off the

stack

and get back to where you were so all

that that context switching is really I

mean there's measures of how people do

this and there's debates about whether

people can can multitask or not whether

they're actually really multitasking or

they're just multi-threading but they're

there's they're dividing the amount of

computational process by the number of

processes you have going so you're

actually not doing multiple things at

once as effectively as if you just stuck
with one so
um yeah anyway and then it's like do we
have that Intel hyper threading capacity
is it a virtualized hyper thread
on one thread
are there actually different threads
yeah I mean the conversation analysis
other than us being right in the thick
of it and it being highly engaging
supported area let's just continue
through this cases here
action observation
okay mirror neurons
this may be very much related to the um
bird song
seeing from the same hymn sheet or
whatever and then like you know doing
the same dance okay attention
this is a super fascinating area
and um
I think a lot of notebooks and and
applications
like just taking a trace of somebody's
browsing on Wikipedia or across websites
and then reconstructing components of
their generative model based upon their
salience revealed choice is a huge area
and then doing again that with the
ocular motor with webcams
uh and doing it with uh
semantic landscapes
hybrid modeling self-organization
arguably with some
pretty interesting advances in the last
several months and days from Fields 11
others
on the

and four stage earlier but on the
morphogenesis as Bayesian inference
okay
in our last
few minutes here
let's just look ahead to nine because I
think this is setting us up well
um
okay
model-based data analysis or databased
modeling or however we want to say this
is about
parameterizing
structured models with observations
so rather than just by Fiat dictate
saying this is the structure of the
model and it spits out observations like
so and so we want to be able to actually
run it backwards not as a generative
model but as a recognition model
so to speak in this expectation
maximization framework and parameterize
our generative model from sparse or
dense observations and that's also
called parametric empirical base
because you're parameterizing the prior
based upon the mean and variance and
patterns of some empirical data
okay the metabazian approach
Bayes theorem
this may even be something
that could be introduced earlier
even though it's also meta
this is like the the map territory
fallacy fallacy
graphical abstract
which is we're making a map here's Us in
the bigger box even though there's a

dashed box even Beyond us
we're making a map
of a cognitive entity
in its map making of its generative
process
and that was the subject also of some
very rich
discussion several days ago with Jacob
and Ali and Dalton
Ollie anything to add on this
uh no nothing much but
um
uh one General comment I wanted to make
about uh chapter nine is that uh you see
because of my project that I've been
working on for the past several months
this chapter nine has been quite helpful
in developing some of my ideas around
how to uh how to exactly model my
um
am I I mean the music emotional
recognition system that I've been
working on but I wanted to point out uh
one small caveat here uh which is that
uh you see at least in my opinion
chapter nine or uh this so-called model
based data analysis uh it's not a
conducive to
generate
every I mean on every model that's been
described throughout the throughout the
book I mean it's just focused on some
specific kinds of models so it might not
be generalizable to every kind of model
that we've seen throughout all the
previous chapters
good point some
models might be just plug and play

so to speak and other ones might take a little bit more work to get to use empirically

okay let's just continue in our last minute so scan through so metabasian process we're drawing inferences about an inferential process

it's a key point and that's the two levels of mapping it's like the map of the map

variational LaPlace

variational Bays

factorized models

for computational Simplicity and interpretability

based upon the sparsity of coupling of variables in a base graph

and LaPlace

using a parabolic approximator of the mode and a variance to ensure

a simple optimizable heuristic

not that it perfectly recapitulates the shape of the posterior which doesn't have to be inverted quadratic

but the LaPlace approximation is smoothly optimizable for both the mode and the variance

and here's where they mentioned that's the parametric empirical Bayesian approach

and basically LaPlace approximation can be thought of as making a commitment to um the quadratic form

so choosing the family that your prior is going to be of and then bootstrapping your initial prior

with your observed data
bootstrapping not in the resampling
sense but in the starting oneself out
sense
um this was explored more
in other
works
but you have some likelihood function of
all the parameters L of data
and
optimizing action
involves Computing beliefs about policy
such that policies are proportionally
taking up slices of the pie
monotonically
with respect to their free energy
and then there's a Precision on action
parameter
where you can turn it all the way down
to sort of like don't know don't care
where even large differences in free
energy between policies get equilibrated
out
that's like hot decision making and then
absolute zero decision making would be a
policy that's even 51 percent more
likely
would be always selected
so
the variational free energy expected
free energy it's abound on surprise
due to Jensen's inequality and evidence
lower bound and all this other work in
variational inference
and then that
pie slicing
can be
tuned up or down with a shaky hand

parameter

soft Max temperature parameter

if the Soft Max is one

the pie is undistorted

if the Soft Max is large

you have a Relentless bias

for better policies

or it could be less than one

here's some technical details of the

LaPlace approximation

parametric empirical base

generalized linear modeling

#spm

here's recipe so this is kind of like

chapter six but now it's model based

data analysis collect behavioral data

maybe the data's already been collected

maybe there's some data sets that we

already already can use

formulate a pmdp

I wonder if this is as chapter six not

as chapter 7.

it could be chapter seven as well if

it's discrete time

specify likelihood

perhaps we can um in the coming weeks

check out SPM mdp vbx

and they mentioned the scripts

data this is nice

this will be fun

chapter seven I mean it is the correct

if they're using this type

four

chapter seven chapter four

so

this looks pretty cool pretty

interesting we could walk through some

examples

false inference

computational pathology

so it kind of ends like chapter eight

with a table of references and a summary

so it's a pretty short chapter

and

has some formalisms

but it's only from page 179 of the PDF

to

196. so it's less than 20 pages

and the next chapter I think is the

longest chapter in the book which one is

chapter 10 chapter 10 197.

yeah that one's over 30 pages

but it ends on a high note

um

all right well

interesting

conversations thanks for this

we will come

um back to chapter nine

look through it and then peep into 10.

go through 10 and really solidify our

understanding

of the map and I guess indirectly of the

territory

and then for those who want to join

will probably continue the discussion

and um especially start looking towards

those project ideas again like we talked

just a little earlier about making a DOT

zero

um

a DOT zero sessions for the textbook

so I I have to apologize I'm I'm

traveling for a few weeks so I'm gonna

miss the next

um several sessions

it's all good thank you Eric

um any other comments people want to

make or I'll close the recording

okay

thank you